What Do YuE's Attention Heads Hear? Probing Musical Characteristics in a Generative Model

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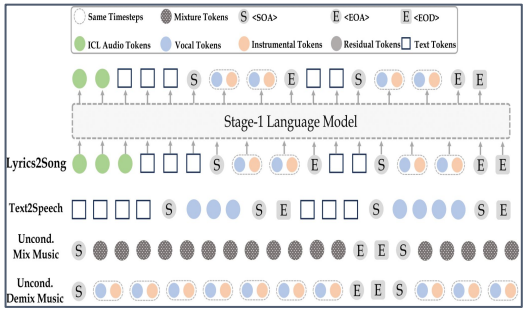
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Motivation

- We investigate genre-specific **attention patterns** revealing insights into YuE's [1] **interpretability** and **bias**
- How do **YuE's attention heads** encode differences between **musical genres**?

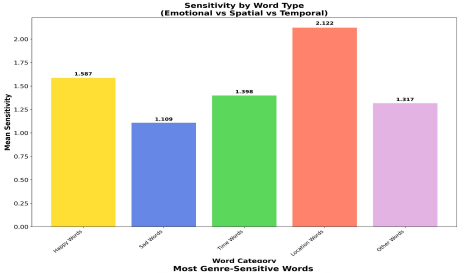
Experiment

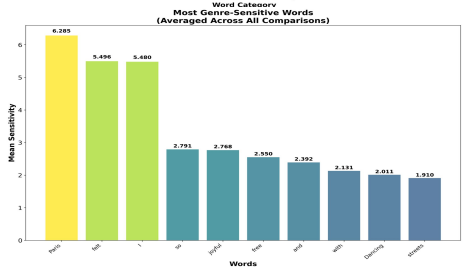
- Used **fixed lyrics** across **4 genres** (Jazz, Metal, Pop, Classical) to isolate genre effects
- Tracked **activations across 32 layers** of YuE-7B Stage 1 during generation
- Compared **layer-level and word-level** activation patterns (e.g., *emotional* vs. *spatial* words)



[1]Yuan, Ruibin, et al. "Yue: Scaling open foundation models for long-form music generation." arXiv preprint arXiv:2503.08638 (2025).

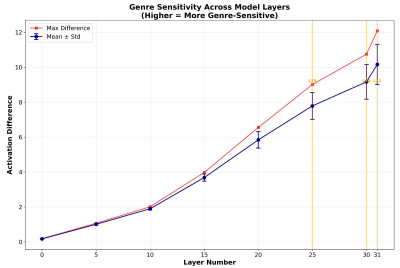
Results

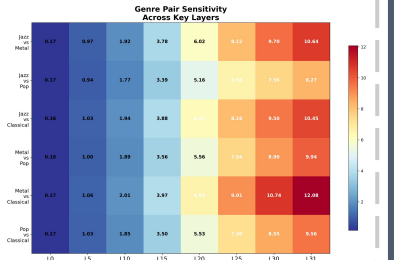




Word Category Analysis Summary
(Most Sensitive Words by Type)

Category	Count	Top Sensitive Words	Avg Sensitivity
Happy	6 words	'free', 'joyful', 'glee'	2.322
Sad	6 words	'melancholy', 'Crying', 'blue'	1.628
Time	5 words	'tomorrow', 'midnight', 'Anci	1.981
Location	6 words	'Paris', 'Tokyo', 'Rome'	2.816





Conclusion & Findings

Layer-level genre sensitivity

- YuE's deeper transformer layers show stronger genre-specific activations
- Pairwise genre comparisons show Metal-Classical has the most distinct neural signature

Word-level genre sensitivity

- Happy and location-based words evoke stronger genre responses than temporal or neutral words, suggesting a semantic-genre link
- *Paris* has the highest mean sensitivity suggesting potential bias

Findings

- The longer the generated music was, the worse the quality got
- Requires constrained prompt to generate stable output
- Single-track ICL faces difficulty with adding extra instruments
- Bias towards certain prompt words such as female voice